TrES-5b

R-0758

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_231106` · created 2026-07-28T03:41:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460254.632 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.047
Transit depth (Rp/Rs)^2	3.23 +/- 1.67 [%]
Orbital Inclination (inc)	81.12 +/- 2.89
Airmass coefficient 1 (a1)	1.041 +/- 0.027
Airmass coefficient 2 (a2)	-0.027 +/- 0.032
Scatter in the residuals of the lightcurve fit is	2.6 %
Best Comparison Star	#2 - [411, 267]
Optimal Aperture	2.81
Optimal Annulus	8.73
Transit Duration (day)	0.045 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.047 vs 0.142 (deviation 0.78σ)
Duration fit vs published	0.045 d vs 0.0758 d (deviation 0.99σ)
Mid-time O–C	9.1 min
Depth SNR	3.7
Residual scatter	2.6 %

Light curve

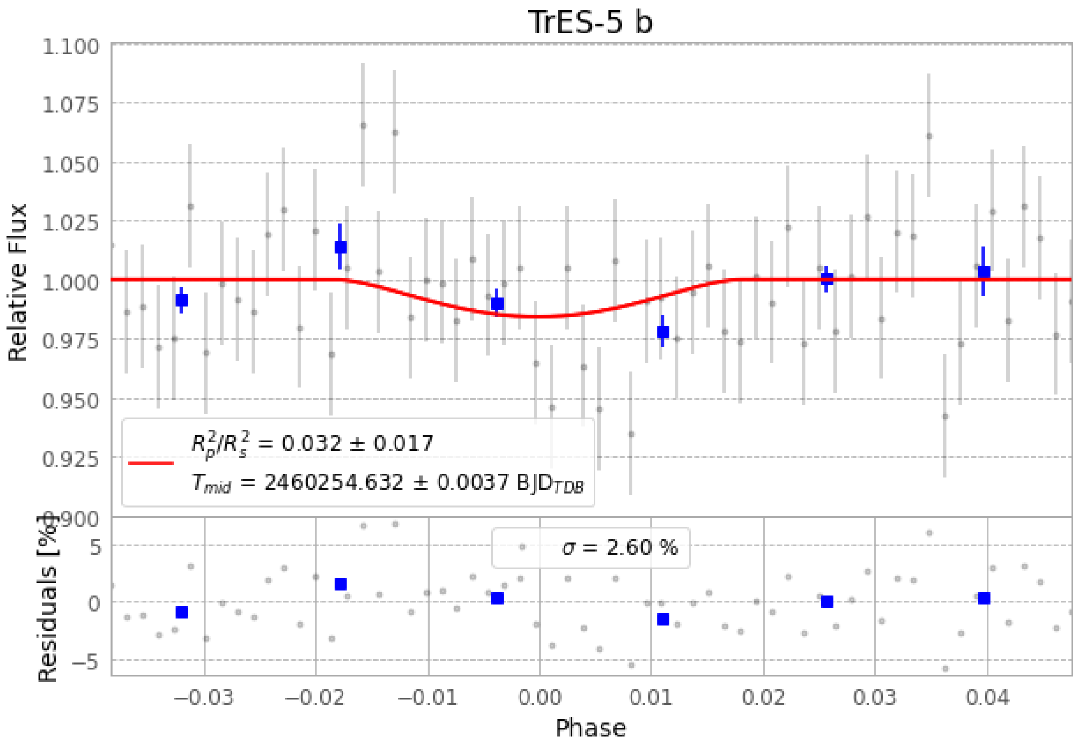

Figure 1 — FinalLightCurve_TrES-5 b_2023-11-05.png (SHA-256 88bba97be7382cca…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 190], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a01ed567924a0086…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

62 FITS frames (41 MB), `TRES-5231106014515.FITS` → `TRES-5231106044815.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-231106.log (edf6fc1b9ab3…), FinalLightCurve_TrES-5 b_2023-11-05.png (88bba97be738…), FinalLightCurve_TrES-5 b_2023-11-05.csv (845fc3e55870…)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)